

1. Title Page

Ahmedabad University
CSE 516 – Probabilistic Graphical Models

Milestone 1: Proposal and Scoping

Bayesian Personalized Recommendation using Latent Trait Models & Expectation Propagation

Category: Theoretical AI

Group 4

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GitHub Repository

https://github.com/Jyotier2006/M1_g4_th_apl_recommend-sys_team

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2. Problem Statement and Motivation

Recommender systems predict which items a user may prefer from a large set of available items. This is difficult because user–item data is usually sparse; a user interacts with only a small fraction of all items.

Our project studies this problem using *Matchbox*, a Bayesian recommendation model developed by Microsoft Research [1]. Matchbox represents users and items using hidden or *latent traits* which are learned from observed ratings and, when available, metadata.

The Bayesian approach is useful because recommendation involves uncertainty. Instead of learning only fixed user and item representations, the model maintains probabilistic beliefs about unknown parameters.

As this is a **Theoretical AI** project, our focus is not simply on building a recommender system. We mainly study the approximate Bayesian inference used in Matchbox, especially Expectation Propagation (EP) and Assumed-Density Filtering (ADF).

3. Research Question and Theoretical Focus

Our main research question is:

How do approximate Bayesian inference choices in the Matchbox latent-trait model, particularly ADF and repeated Expectation Propagation, affect predictive quality and computational cost?

A simplified latent preference between user u and item i can be written as

$$a_{ui} = \mathbf{s}_u^T \mathbf{t}_i + b_u + b_i + \epsilon,$$

where $\mathbf{s}_u$ and $\mathbf{t}_i$ represent latent user and item traits, b_u and b_i represent bias terms, and ϵ represents uncertainty.

The main theoretical interest is the trade-off between faster approximate inference and improved posterior/predictive quality.

4. Scope and Assumptions

The current project scope includes:

- Bayesian latent user and item traits,
- explicit 1–5 star rating prediction,
- ADF and repeated EP inference,
- different latent dimensions,
- metadata versus no metadata, and
- increasing sparsity / cold-start conditions.

Our main assumptions are:

- User preferences and item characteristics can be represented using lower-dimensional latent traits.
- Observed ratings provide evidence about these hidden traits.
- The graphical structure is taken from Matchbox; therefore, we focus mainly on **parameter learning and inference**, not structure learning.
- The same experimental conditions will be maintained while comparing different methods.

5. Base Paper and SOTA Positioning

Base Paper – Matchbox, Microsoft Research (2009)

Our base paper is *Matchbox: Large Scale Online Bayesian Recommendations* by Stern, Herbrich and Graepel [1].

The paper models users and items in a low-dimensional latent trait space and uses Bayesian inference to learn hidden preferences from observed feedback. It combines Expectation Propagation and Variational Message Passing, while ADF allows efficient single-pass online learning.

The paper evaluates the model using MovieLens and Netflix data. One important observation for our project is that additional EP passes can improve predictions, but require additional computation.

This makes Matchbox directly suitable for our theoretical study of approximate inference.

Recent SOTA Anchor – PreRec, AWS AI Labs (2024)

Our recent SOTA reference is *Pre-trained Recommender Systems: A Causal Debiasing Perspective* [2], published at WSDM 2024 with authors from AWS AI Labs.

PreRec uses a hierarchical Bayesian recommendation framework and focuses on recommendation in new domains with limited interaction data.

It is not an EP-based replacement for Matchbox. We use it as a recent industry research reference to understand how modern Bayesian recommendation has evolved, especially for sparse and limited-data settings.

6. Dataset and Benchmark

We will use the **MovieLens 1M** dataset released by GroupLens Research [3].

It contains approximately:

- one million ratings,
- 6,000 users,
- 4,000 movies, and
- ratings on a 1–5 scale.

The main rating fields are:

User ID, Movie ID, Rating, Timestamp.

The dataset also contains user information such as age group, gender and occupation, and movie information such as genres. This makes it suitable for our metadata-versus-no-metadata experiment.

MovieLens is particularly relevant because it was also used in the original Matchbox evaluation.

Dataset source:

<https://grouplens.org/datasets/movielens/1m/>

7. PGM Framework

Our project covers all three required PGM components: **Representation, Learning and Inference.**

7.1 Representation

The major variables are:

- latent user traits S_u ,

- latent item traits T_i ,
- user and item biases,
- latent user–item affinity,
- observed rating R_{ui} ,
- rating thresholds, and
- optional user/item metadata.

The observed rating depends on hidden user and item characteristics. This is naturally represented using a probabilistic factor graph because the model contains several interacting latent and observed variables.

7.2 Learning

The graphical structure is fixed. We therefore focus on **Bayesian parameter learning**.

From observed ratings, the model learns posterior beliefs about latent user traits, item traits, biases and other model parameters.

7.3 Inference

For an unseen user–movie pair, the main inference query is

$$p(R_{ui} \mid \text{observed ratings and metadata}).$$

Exact posterior computation is difficult for a large latent model. Matchbox therefore uses approximate message passing.

Our main comparison will be:

$$\text{ADF (single pass)} \quad \text{vs.} \quad \text{multiple EP updates.}$$

8. Baselines

We plan to use the following baselines:

1. **Mean-based predictor** – simple global/user/item mean rating prediction.
2. **Matrix Factorization / SVD-style recommender** – conventional non-Bayesian latent-factor baseline.
3. **Single-pass ADF** – inference baseline for comparison with repeated EP updates.

The intended comparison is:

$$\text{Baseline} \rightarrow \text{Matchbox PGM} \rightarrow \text{SOTA Positioning.}$$

9. Initial Key Performance Indicators

Our initial evaluation measures are:

KPI	Purpose
MAE / RMSE	Measure rating-prediction error.
Predictive log-likelihood / NLL	Measure probabilistic model quality.
Inference / training time	Measure computational cost.
Number of EP iterations	Measure convergence and inference effort.
Calibration, where feasible	Study whether predictive uncertainty is reliable.

Experiments will use multiple random seeds or splits where appropriate, and final results will be reported as mean $\pm$ standard deviation.

10. Experimental Plan

The planned experiments are:

Exp.	Study	Purpose
E1	Baseline setup	Prepare MovieLens and establish initial baseline performance.
E2	ADF vs. repeated EP	Study prediction improvement against computational cost.
E3	Latent dimensionality	Study model capacity, prediction error and runtime.
E4	Metadata vs. no metadata	Measure the effect of user/movie side information.
E5	Sparsity / cold start	Study behaviour when less rating information is available.

At M1, these experiments are the **planned research direction**. No unperformed experimental result is claimed in this report.

11. Team Responsibilities

The initial responsibility distribution is:

Member	Primary Responsibility
Jyotier Vithalani	Matchbox mathematical formulation, EP/ADF study and repository co-ordination.
Tanmay Kanani	Baseline methods and implementation planning.
Ansh Chaudhari	Base-paper and recent SOTA literature analysis.
Shlok Shah	MovieLens preparation and metadata study
Het Patel	KPI design, experiment tracking and documentation.

The responsibilities are primary assignments only. All members will study the complete graphical model, dataset, learning, inference and evaluation process.

12. Timeline

The project will be carried out according to the four course milestones.

Milestone	Date	Planned Work
M1	15 Sep 2026	Problem, mathematical question, assumptions, benchmark, PGM mapping, baselines, SOTA, KPIs and experiment plan.
M2	11 Oct 2026	Mathematical formulation, preprocessing, implementation, baseline setup and initial experiments.
M3	1 Nov 2026	Full experiments, robustness study, benchmark comparison and statistical analysis.
M4	22 Nov 2026	Final validation, KPI plots/tables, limitations, reproducibility and final SOTA positioning.

13. GitHub Repository

The project repository is:

https://github.com/Jyotier2006/M1_g4_th_apl_recommend-sys_team

For Milestone 1, the repository will contain the report, technical walkthrough, dataset information and any available technical artifacts.

The M1 folder will follow the required structure:

```
M1_g4_th_apl_recommend-sys_team/  
|-- README.md  
|-- Report/  
|-- Video/  
|-- Data/  
|-- Code/          (if applicable)  
'-- Results/       (if applicable)
```

14. References

References

- [1] D. H. Stern, R. Herbrich, and T. Graepel, “Matchbox: Large Scale Online Bayesian Recommendations,” in *Proceedings of the 18th International Conference on World Wide Web (WWW '09)*, 2009, pp. 111–120. doi: 10.1145/1526709.1526725.
<https://dl.acm.org/doi/10.1145/1526709.1526725>
- [2] Z. Lin, H. Ding, N. T. Hoang, B. Kveton, A. Deoras, and H. Wang, “Pre-trained Recommender Systems: A Causal Debiasing Perspective,” in *Proceedings of the 17th ACM International Conference on Web Search and Data Mining (WSDM '24)*, 2024, pp. 424–433. doi: 10.1145/3616855.3635779.
<https://doi.org/10.1145/3616855.3635779>
- [3] GroupLens Research, “MovieLens 1M Dataset.” [Online]. Available: <https://grouplens.org/datasets/movielens/1m/>